

Impact of Additional Disabilities on Adaptive Behavior and Support Profiles for People With Intellectual Disabilities

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Abstract

Numerous researchers have reported a high incidence of additional disabilities coexisting with intellectual disabilities. Although an intuitive link can be made between the existence of multiple disabilities and greater need for support, little has been reported about this relationship. Using measures of adaptive functioning and support needs, we examined the extent to which adaptive and challenging behaviors and consequent support needs (including medical) were impacted by the presence and severity of additional disabilities for individuals with intellectual disabilities. Results show that adaptive behaviors and support needs were meaningfully related to the number and severity of additional disabilities present, whereas this was not so for challenging behaviors. Findings are discussed in terms of contemporary models of disability and functioning.

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There is widespread recognition of the extent of multiple disabilities in individuals with primary intellectual disabilities (Australian Institute of Health and Welfare, 2003; McLaren & Bryson, 1987; Murphy, Boyle, Schendel, Decouflé, & Yeargin-Allsopp, 1998; Rohde, Farmer, Glover, & Nursey, 1989), although comparisons between studies are complicated by different methodological approaches. Thus, according to the Australian Institute of Health and Welfare (2007), 52% of all recipients of commonwealth and state government-funded services with a primary intellectual disability also report the presence of an additional disability. This figure rises to 73% when considering only recipients of funded accommodation support services (Australian Institute of Health and Welfare, 2007). However, estimates of the prevalence of multiple disabilities in the literature have differed considerably from the Australian data source, ranging from 22% (Murphy, Yeargin-Allsopp, Decouflé,

& Drews, 1995) to 58% (van Isterdael, Stilma, Bezemer, & Tijmes, 2006).

Methodological issues considered to be of significance in terms of contributing to variances in prevalence rates include definitional issues involving different criteria for inclusion in studies; the diagnostic methods used (i.e., reviews of case notes, assessment with screening instruments, or clinical assessment); and the range of additional disabilities examined, with some researchers focusing only on the presence of a small and specific range of disabilities (Kapell et al., 1998; McLaren & Bryson, 1987; Morgan, Baxter, & Kerr, 2003; Murphy et al., 1995; Smiley, 2005; Whitaker & Read, 2006). Level of intellectual functioning is also considered to be important, with several authors reporting substantially higher prevalence rates of additional disabilities for individuals with more marked levels of intellectual impairment (Australian Institute of Health and Welfare, 2003; Christianson et al., 2002;

Drews, Yeargin-Allsopp, Decouflé, & Murphy, 1995; Gustavson, Umb-Carlsson, & Sonnander, 2005; McLaren & Bryson, 1987; Murphy et al., 1995). This is considered to reflect some underlying organic cause associated with more severe levels of intellectual disability (Lewis et al., 2000; Murphy et al., 1998).

Disabilities reported to coexist with intellectual disabilities have included physical, neurological, sensory, and psychiatric disabilities. Each has been reported to occur at a rate higher than in the general population (Carvill, 2001; McGrother et al., 2006; van Schrojenstein Lantman-de Valk, 2005; White, Chant, Edwards, Townsend, & Waghorn, 2002). In Australia, the Institute of Health and Welfare (2003) reported prevalence rates of these coexisting disabilities to be 25% for physical/diverse disabilities; 29% for sensory disabilities (vision, 2%; hearing, 5%; and speech, 26%); and 31%, for psychiatric disabilities, when the primary disability is intellectual.

In addition to higher incidence rates, each of these categories has been reported to be more prevalent for individuals with more marked levels of intellectual impairment (Christianson et al., 2002; Evenhuis, Theunissen, Denkers, Verschuure, & Kemme, 2001; Jacobson & Janicki, 1983; McDermott et al., 2005; McGrother et al., 2006; Murphy et al., 1995; Warburg, 2001; Zijlstra & Vlaskamp, 2005), although there have been conflicting results reported regarding the relationship between prevalence of psychiatric disabilities and severity of intellectual impairment. Thus, although an increase in prevalence of psychiatric disabilities with increased severity of intellectual impairment has been reported (Cooper, Smiley, Morrison, Williamson, & Allan, 2007; Whitaker & Read, 2006), others have found the opposite pattern (Crews, Bonaventura, & Rowe, 1994; Holden & Gitlesen, 2004; Iverson & Fox, 1989; La Malfa, Notarelli, Hardoy, Bertelli, & Cabras, 1997; Slone, Durrheim, Kaminer, & Lachman, 1999); and yet other researchers have reported no increased risk of psychiatric disability associated with degree of intellectual impairment (Stromme & Diseth, 2000). Recently, researchers have looked to psychiatric disorders as a possible explanation for challenging behaviors associated with intellectual disabilities, finding that the relationship is by no means straightforward (Holden & Gitlesen, 2004).

Although the different methodological issues raised earlier are considered significant in leading

to the discrepancies between studies, there is an indication that diagnostic difficulties contribute to the underrecognition of psychiatric disorders for people with intellectual disabilities (Kosken-Tausta, Iivanainen, & Almqvist, 2002; White et al., 2002). Similarly, health impairments of individuals with intellectual disabilities are also frequently underrecognized (Gustavson et al., 2005). Communication problems are considered to be significant in the underreporting of health and other impairment conditions. Thus, limited communication skills restrict capacity to verbalize symptoms and communicate pain or discomfort, and caregiver detection becomes the primary means of impairment identification. However, in a review of medical needs for individuals with intellectual disabilities, Kerr et al. (2003) reported that despite having health/medical qualifications, nurse caregivers frequently failed to detect vision and hearing loss in individuals with intellectual disabilities; and they suggested that other health conditions may also be overlooked. Yet, as with the additional disabilities already discussed, these health conditions are considered to be more prevalent for individuals with intellectual disabilities than in the general population. The functional significance of these chronic health problems has been reported to be more severe for individuals with existing impairments (Traci, Seekins, Szalda-Petree, & Ravesloot, 2002) and comorbidity of long-term health conditions has been found to be highly correlated with severity of disability (Australian Institute of Health and Welfare, 2004).

Contemporary models of functioning and disability, such as the *International Classification of Functioning, Disability and Health–ICF* (World Health Organization, 2001), provide a framework for understanding the association between an individual's health condition or impairments and the resultant activity limitations and participation restrictions. Intrinsic to this model is an interaction between the individual's health condition or impairments and environmental factors, which presents barriers to full participation in everyday activities. Although the high incidence of health impairments and additional disabilities has been generally reported for individuals with intellectual disabilities, little is known about the extent to which these additional impairments and disabilities impact on functional capacity and consequent need for support. An intuitive connection can be made between the presence of multiple

disabilities, reduced functional capacity, and the need for greater supports. Understanding the impact of additional disabilities and their severity on need for supports is likely to have important implications for service development and resource allocation, both in terms of the level of support funds required and the types and levels of caregiver expertise. According to Rohde et al. (1989), a person's support and resource needs will be influenced as much by the presence of additional disabilities and health concerns as by the presence of an intellectual impairment. Consistent with this, Carvill (2001) suggested that the presence of coexisting hearing and intellectual disabilities are likely to have a multiplicative rather than an additive cumulative impact on an individual's cognitive capacity.

Severity of intellectual impairment has been reported to be associated with level of adaptive skills. Correlations between scores obtained on adaptive behavior and intelligence scales vary according to the type of scale used and across samples; coefficients are generally in the moderate range, with lower correlations found with challenging behavior measures (Harrison, 1987; Meyers, Nihira, & Zetlin, 1979). Although severity of intellectual impairment has been examined with respect to adaptive behaviors, it has not been investigated with regard to need for support or severity of the person's total disability due to the presence of additional disabilities.

In the present study we investigated the impact of indicators of disability severity on adaptive and challenging behaviors and need for supports (including supports for medical/health and behavioral problems) for individuals with a primary intellectual disability. Indicators of disability severity included the number of disabilities present, the severity of the primary disability, and a cumulative measure of disability severity that incorporates the rated severity of each additional disability present. We anticipated finding fewer adaptive behavior skills, coupled with a need for more support, associated with having more disabilities and/or with more severe levels of disabilities. However, what was not apparent based on previous research was whether the presence of additional disabilities would produce a profile indicating a steady change in skill levels and associated levels of need or whether some other trend would arise to explain the impact of the presence of additional disabilities for people with intellectual disabilities. We anticipated that a

trend of greater need for behavioral support associated with having more disabilities and/or more severe levels of disabilities was less likely, given prevalence profiles and lower correlations between challenging behaviors and severity of intellectual impairment reported in the literature.

Method

Participants

Participants were 83 individuals (28 females, 54 males, one gender not reported) with intellectual disabilities receiving support from one of five participating South Australian residential facilities. They ranged in age from 20 to 72 years ($M = 40.9$, $SD = 10.7$). Classifications for severity of intellectual impairment and the nature and severity of additional disabilities were provided by residential caregivers on the basis of their knowledge of the person and/or through reference to individual client records. Severity of intellectual impairment classifications provided were mild ($n = 6$, 7%), moderate ($n = 49$, 59%), severe ($n = 25$, 30%), and profound ($n = 3$, 4%). Information regarding the presence of additional disabilities was collected from caregivers involved in the study who used the disability item taken from the National Minimum Data Set Service User Form developed by the Australian Institute for Health and Welfare (2002) at the same time that the adaptive behavior and support need scales were completed. This form was designed to collect critical information on all users of government-funded services for the purposes of planning and evaluation of funded programs. It provides a checklist for recording primary and any other significant disabilities causing difficulties for service users and has accompanying descriptive information on the disability categories included on the form to guide classification. The total number of disabilities was obtained by summing the number of disability categories into which each individual was placed. Information about the severity of the additional disabilities was obtained using a severity rating scale (mild, moderate, severe, or profound), where severity was considered in terms of the impact on the person's life in an attempt to obtain some consistency with respect to degree of severity across different disability categories. Single and coexisting disabilities reported were as follows: one disability ($n = 19$, 23%), two disabilities ($n = 23$, 28%), three

disabilities ($n = 27$, 32%), four disabilities ($n = 12$, 14%), and five disabilities ($n = 2$, 2%). Thirty nine (71%) of the 55 people with mild/moderate intellectual impairment and 25 (89%) of the 28 people with a severe/profound level of impairment had one or more additional disabilities.

Instruments

Inventory of Client and Agency Planning (ICAP). The ICAP (Bruininks, Hill, Weatherman, & Woodcock, 1986) was designed to assess the adaptive and maladaptive behaviors of individuals with intellectual and developmental disabilities. It is a standardized instrument that provides separate scores for adaptive and maladaptive (or challenging) behaviors. The adaptive areas assessed by ICAP are Motor Skills, Social and Communication Skills, Personal Living Skills, and Community Living Skills. Scores on these four subscales are used to derive an overall indicator of adaptive behavior, the Broad Independence Index, with higher scores on this index indicating more adaptive behavior skills. The frequency and severity of eight problem behavior areas assessed with the ICAP result in ratings used to establish four indices of maladaptive behavior (Internalized, Asocial, Externalized, and Generalized). The more negative the score on the maladaptive behavior indices, the more severe the problem behaviors. This instrument also generates a Service Score, which is a combined measure of adaptive and maladaptive behaviors that indicates the overall level of supervision, care, or training that an individual requires, that ranges from 0 to 100, with higher scores indicating greater degrees of independence and, consequently, less need for support. The ICAP manual reports good reliability coefficients for all measures of adaptive behavior (ranging from .74 to .98 for test–retest and .72 to .94 for interrater) and challenging behavior (ranging from .74 to .92 for test–retest and .69 to .81 for interrater) derived from a number of reliability studies in addition to discussion of various studies that support the validity of the instrument.

Service Need Assessment Profile–SNAP. This assessment instrument was developed by Gould (1998) to measure support needs of individuals with different disability types and levels of severity who were receiving accommodation and day support services. The scale measures 29 areas of support across five subscales: Personal Care (i.e.,

bathing, dressing, eating, meal preparation, household tasks, and self-protective skills); Physical Health (i.e., ambulation, health issues, incontinence, mobility, pressure care, and epilepsy); Behavior Support (covering type of behavior, support requirements, behavioral risk, behavioral programs, and mental health issues); Night Support (i.e., safe practices, sleeping patterns, physical support, health monitoring, and behavioral issue); and Social Support (i.e., communication, social, leisure, money skills, day support requirements, skill development options, and travel needs). Each item must be rated, with ratings generally ranging from a score of 1 (*independent for activities of daily living*) to 5 (*totally dependent on caregiver support for activities of daily living*). Computer software generates a profile that includes an allocated time, in hours per day, for providing support to the individual for each of the five support subscales as well as a separate day support hour total score, with higher scores indicating greater need for support.

Results of validity studies for the SNAP provide support for the use of this instrument as a support need measure (Guscia, Harries, Kirby, Nettelbeck, & Taplin, 2006); reliabilities for the five SNAP subscales have been reported to range from .86 to .97 for test–retest reliability and from .61 to .91 for interrater reliability (Guscia, Harries, Kirby, Nettelbeck, & Taplin, 2005).

Supports Intensity Scale–SIS. The SIS (Thompson et al., 2004) is a support need instrument designed to measure the intensity of supports required by individuals with intellectual and developmental disabilities for performance of everyday activities. The SIS is comprised of three sections. Section 1 (Support Need Scale) assesses supports needed in six areas: Home Living, Community Living, Life-Long Learning, Employment, Health and Safety, and Social Activities. Assessors using the SIS are asked to indicate the frequency of support required, daily support time, and type of support to enable the individual to participate successfully in the activity listed. A Support Need Index (i.e., the composite standard score) is derived using the six Section 1 subscales, with higher scores on the subscales and Support Need Index representing greater need for support. Section 2 is a supplementary scale that addresses supports in the area of Protection and Advocacy. It is scored in the same manner as the Section 1 subscales. Section 3 is used to measure Exceptional Medical and Behavioral Support Needs with the

presence of exceptional medical conditions (16 items) and problem behaviors (13 items) that are rated using a scale of 0 (*no support needed*), 1 (*some support needed*) or 2 (*extensive support needed*). The SIS manual provides support for the content, criterion-related, and construct validity of the SIS. Reliabilities for the Section 1 subscales are reported in the manual to range from .52 to .82 for test–retest reliability. Interrater reliability for the Section 1 subscales are reported to range from .47 to .92 (Thompson, Tassé, & McLaughlin, 2008).

Procedure

Using ICAP, SNAP, and SIS, caregivers assessed residents with whom they had worked closely for a minimum of 3 months. Assessments were presented in random order to address the possibility that information sought on one assessment may alter information provided on subsequently presented scales. Administration guidelines provided in manuals were followed for each instrument.

Data Analysis

Indicators of disability severity used in this study included severity of intellectual impairment, the total number of disabilities present, and a cumulative measure of total disability severity. This measure, the Cumulative Severity Score (CSS), was derived using a crude and exploratory method that involves the addition of severity ratings provided by caregivers for the primary and each additional disability, with mild = 1, moderate = 2, severe = 3, and profound = 4. The CSS for study participants ranged from 1 to 14 (mean = 5.7, *SD* = 3.2).

Measures investigated in this study included the ICAP Broad Independence Index as a measure of adaptive behavior. Challenging behavior was examined using one measure of challenging behaviors present (ICAP Generalized Maladaptive Index) and two measures of support required for behavior (SIS Exceptional Behavior Needs Total Score and SNAP Behavior Support Hours total). To assist in understanding the profiles obtained, we have reported the ICAP Generalized Maladaptive Index as an absolute value, given that positive values present in the data set for this item were zero. Consequently, higher values on this variable reflect the presence of more challenging behaviors. Overall need for support was examined using three measures: ICAP Service Score, SIS Support

Need Index, and SNAP Day Support Hours total. Two measures of need for medical/health supports were considered: the SIS Exceptional Medical Needs total score and SNAP Physical Health Support Hours total.

One of our aims in this study was to investigate the impact of having more disabilities and/or more severe disabilities on profiles of functioning and need for support. Thus, we graphed the profiles produced for the number and overall severity of disabilities for each of the measures, using Z scores for graphing due to wide variations in score ranges across the different instruments. Regression curve estimation procedures were used to examine the trends present in the data. Using these procedures, it is possible to test the goodness of fit for a number of models. However, our primary interest here was to determine whether there was a trend of fewer skills and increased need for support associated with having more disabilities and/or more severe levels of disabilities. Therefore, rather than testing for models in which it was possible to have a change in trend directions (e.g., quadratic or cubic), we decided to investigate only the goodness of fit for three models. In particular, we tested for the presence of a linear trend (i.e., a steady rise in level of need or fewer adaptive skills as disability severity increases), an exponential trend (i.e., rate of change in skill/need scores increases as disability severity increases), or a logarithmic trend (i.e., rate of change slows as disability severity increases). It was not possible to check for the presence of an exponential trend when the data contained zero values. Consequently, we made adjustments to variables when required by adding one to all data items to remove zero values, while maintaining the data trend. Curve estimation for variables with distributions that were not normally distributed were conducted for both transformed and untransformed data, to check the results obtained.

Finally, we conducted analysis of variance (ANOVA) with planned comparisons and Kruskal-Wallis and Mann-Whitney *U* tests to examine the impact of having more disabilities present and/or more severe levels of disabilities. One-tailed tests of significance were employed for all measures, except for the three challenging behavior measures, owing to the directional nature of our hypotheses for these variables (i.e., fewer skills and need for more support associated with having more disabilities present and/or more severe levels of disabili-

ties). However, we did not anticipate such directional associations for challenging behavior and for these, two-tailed tests were applied.

Results

Prevalence Rates

Consistent with data from the Australian Institute of Health and Welfare (2007) for accommodation support services, a large number of participants ($n = 64$, 77%) had one or more additional disabilities. The relationship between number of disabilities present and severity of intellectual disability was significant, Spearman's $\rho .33$, $p < .01$. Table 1 reports the prevalence of additional disabilities for individuals with mild/moderate and severe/profound intellectual disabilities and for the total group. Comparing prevalence rates of additional disabilities between the two severity groups, we found significantly higher prevalence rates for the presence of physical disabilities, $\chi^2(1, N = 83) = 4.00$, $p = .05$, and speech disabilities, $\chi^2(1, N = 83) = 21.31$, $p < .001$, for the severe/profound impairment group. Prevalence differences between groups were not significant for neurological and sensory disabilities and could not be established for the remaining disability groups (autism, acquired brain injury, vision, hearing, and psychiatric) due to low incidence rates in the study group. We also compared prevalence rates of

medical issues for the two severity groups. On the basis of whether any medical issues were indicated in the SIS Exceptional Medical Supports scale (i.e., a score > 0 , which indicated that at least some support was required for medical issues), a significant difference was obtained, $\chi^2(1, N = 83) = 6.51$, $p = .01$, with a higher prevalence for the severe/profound group ($n = 20$, 71%) compared with the mild/moderate group ($n = 23$, 42%).

Relationships Between Severity of Disability, Adaptive and Challenging Behaviors, and Support Needs

To investigate the impact of disability severity on adaptive and challenging behaviors and need for support, we correlated the indicators of disability severity with the scale scores. Additionally, due to differences in disability prevalence rates described above, we also conducted partial correlations for number of disabilities present, controlling for severity of intellectual impairment. Results are presented in Table 2. Significant correlations were obtained between each of the indicators of disability severity and scores obtained on measures of adaptive behavior, support needs, and medical needs. By and large, there was no such association between disability severity indicators and the presence of challenging behaviors or need for behavioral support.

Table 1. Number of Associated Disabilities Present by Level of Intellectual Disability

Additional disabilities present	Level of intellectual disability				Total (<i>N</i> = 83)	
	Mild/moderate (<i>n</i> = 55)		Severe/profound (<i>n</i> = 28)			
	<i>n</i>	%	<i>n</i>	%	<i>N</i>	%
Autism	3	6	1	4	4	5
Physical	12	22	12	43	24	29
Acquired brain injury	1	2	0	0	1	1
Neurological	12	22	11	39	23	28
Sensory ^a	13	24	5	19	18	22
Vision	9	16	4	14	13	16
Hearing	5	9	1	4	6	7
Speech	14	26	22	79	36	43
Psychiatric	12	22	2	7	14	17

^aTotal does not equal the sum of the vision and hearing prevalence rates due to coexistence of sensory impairments for one individual.

Table 2. Indicators of Correlations Between Measures of Severity and Adaptive and Challenging Behavior and Support

Measure ^a	Indicators of disability severity			
	ID ^b severity ^d	Disability no. ^d	Disability no. (ID severity controlled)	CSS ^{c,d}
Adaptive behavior				
ICAP Broad Independence Index	−.68**	−.61**	−.51**	−.74**
Challenging behavior				
ICAP Generalized Maladaptive Behavior ^e	−.24*	−.01	.19	−.16
SIS Exceptional Behavior Needs Total	−.03	.01	.07	.02
SNAP Behavior Support Hours	−.09	−.10	−.10	−.16
Support needs				
ICAP Service Score	−.58**	−.60**	−.52**	−.68**
SIS Support Need Index	.71**	.59**	.49**	.71**
SNAP Day Support Hours	.51**	.51**	.43**	.54**
Medical/health needs				
SIS Exceptional Medical Needs Total	.29**	.50**	.47**	.48**
SNAP Physical Health Support Hours	.47**	.68**	.60**	.70**

^aICAP = Inventory of Client and Agency Planning, SIS = Supports Intensity Scale, SNAP = Service Need Assessment Profile. ^bIntellectual disability. ^cCumulative Severity Score. ^dSpearman rho correlations. ^eAbsolute values used.

* $p < .05$. ** $p < .01$.

In subsequent analyses we addressed the impact of two indicators of disability severity on skill levels and need for support: number of disabilities present and the CSS.

Trends for the Number of Disabilities Present and Severity of Disabilities

Figures 1, 2, and 3 graphically depict the nature of the trends linking the number of disabilities and the severity of disabilities for the measures of skills, behavior, and support need. The patterns of results in Figure 1 indicate that as the number and cumulative severity of disabilities increases, fewer adaptive skills are present (indicated by the negative slope for ICAP Broad Independence function as the number and severity of disabilities increase). The need for more overall support (Figure 1) associated with having more disabilities and more severe levels of disabilities is evident by the positive slope for functions describing both the SIS Support Need

Index and SNAP Day Support Hours as well as the negative slope of the ICAP Service Score (where lower scores represent a greater need for support and care). Additionally, Figure 2 indicates that with an increase in the number of disabilities present and severity of disabilities, more support is required for medical/health issues, as evident by the positive slopes of functions for both the SIS Exceptional Medical Needs total score and SNAP Physical Health Support Hours measures.

Trends, however, were less straightforward for the challenging behavior measures (Figure 3), for both the number of disabilities present and the CSS. Although the two measures of need for behavioral support (SNAP Behavior Support Hours and SIS Exceptional Behavior Need total) reflect a need for greater support, consistent with the presence of more challenging behaviors (ICAP Generalized Maladaptive Behavior), there are a number of directional changes associated with

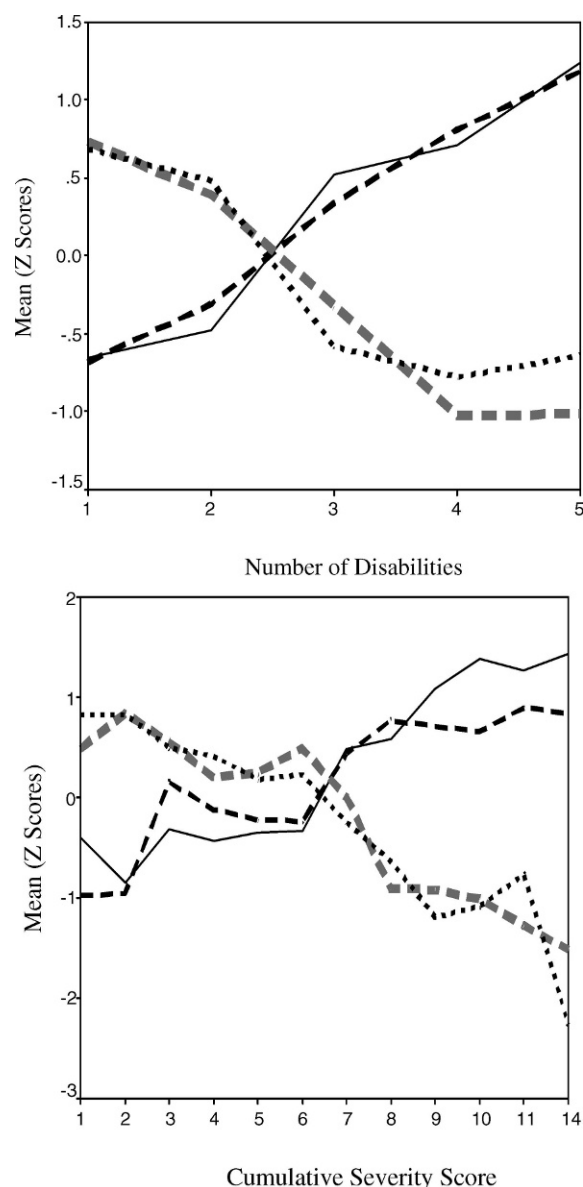


Figure 1. Graphs of adaptive behavior and overall support needs scores for the number of disabilities present (top graph) and the Cumulative Severity Score (bottom graph). Lines represent scores for ICAP Broad Independence (short dashes, black line), ICAP Service Score (long dashes, gray line), SIS Support Need Index (solid black line), and SNAP Day Support Hours (long dashes, black line).

having more disabilities present and more severe levels of disabilities. Thus, overall, this suggests no clear cut linear relationship with these two indicators of disability severity, as was evident from the correlational results.

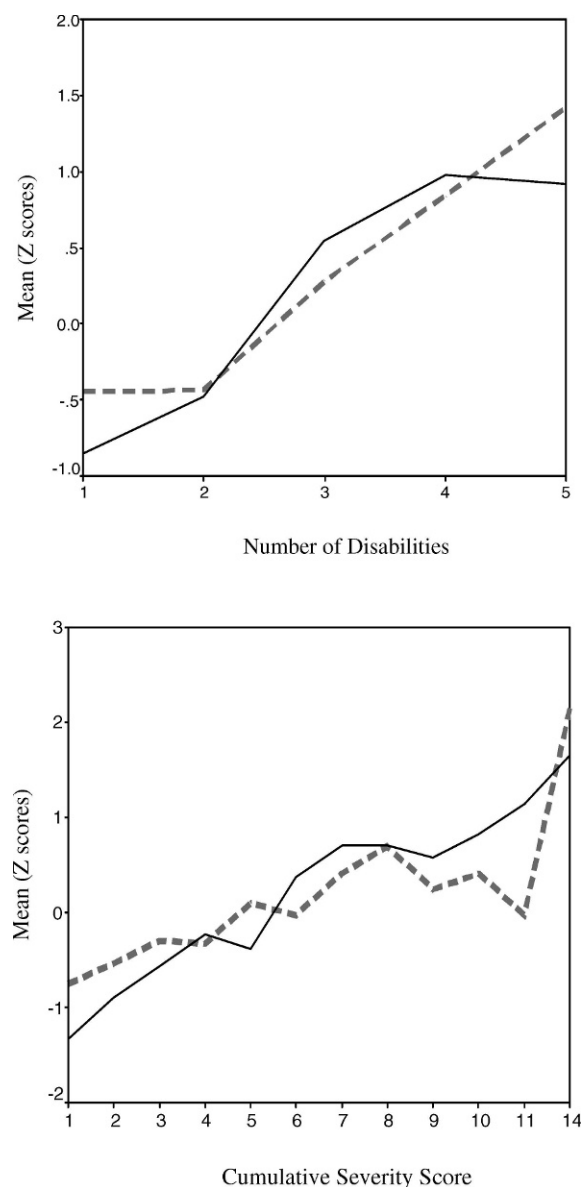


Figure 2. Graphs of medical/health support indicators for the number of disabilities present (top graph); and the Cumulative Severity Score (bottom graph). Lines represent scores for SNAP Physical Health Support Hours (solid black line) and SIS Exceptional Medical Needs Total (long dashes, gray line).

We used curve estimation to generate regression statistics to examine the trends present in the data. However, the results of these analyses were not informative, with no clear cut trends identified. For all measures except the challenging behavior scales, we obtained very similar and significant R^2 values for each fit for the three

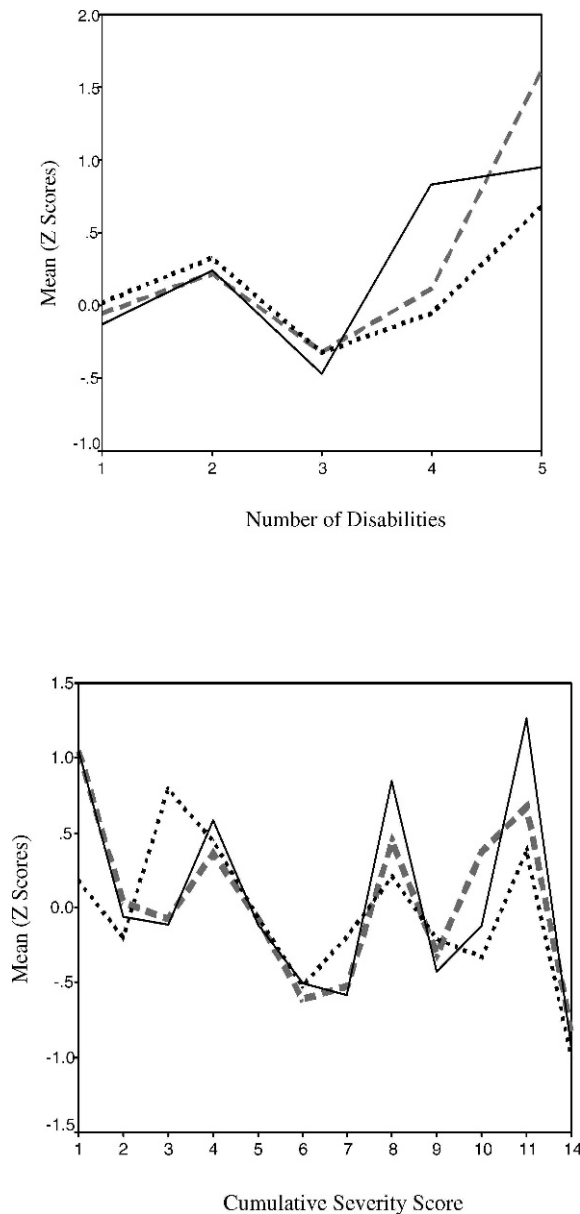


Figure 3. Graphs of challenging behavior and behavioral support indicators for the number of disabilities present (top graph) and the Cumulative Severity Score (bottom graph). Lines represent scores for ICAP Generalized Maladaptive Behavior (solid black line), SNAP Behavior Support Hours (short dashes, black line), and SIS Exceptional Behavior Needs Total (long dashes, gray line).

models compared (i.e., linear, exponential, and logarithmic), when considering the two indicators of disability severity (i.e., number of disabilities present and the CSS). Values of R^2 s ranged from .21 to .43 (median = .32) for number of

disabilities and .22 to .61 (median = .45) for CSS. Only small differences were obtained between R^2 values for each of the three trend models compared for each of the need/skill measures, making it difficult to determine the predominant trends present in the data when considering these two indicators of disability severity. In summary, however, the trends present in Figures 1 and 2 and the significant R^2 values obtained in curve estimation analyses do provide support for the conclusion that higher levels of need and fewer skills are associated with having more disabilities and more severe levels of disabilities. However, determination of whether the rate of change remains steady, slows, or increases as the number and severity of disabilities increases was not possible with these data.

The pattern of results for the measures of challenging behavior was different. No significant model fits were found for the number of disabilities present. Two very small R^2 values (both .05) were obtained for model fits for an exponential trend for the CSS for SNAP Behavior Support Hours (transformed) and ICAP Generalized Maladaptive Behavior Index (transformed); but these results were not supported for curve estimation conducted with the untransformed variables. There was no support for the presence of linear or logarithmic growth trends between having more disabilities or more severe levels of disabilities and measures of challenging behavior.

Impact of Number and Severity of Disabilities on Adaptive and Challenging Behaviors and Support Needs

To examine differences between sequential groups, we used one-way ANOVAs and conducted planned comparisons or, as required, nonparametric analyses using Kruskal-Wallis and Mann-Whitney U tests to establish whether scores on the measures used in the study differed significantly according to reported number of disabilities present or on the basis of CSS.

The results of these analyses and the mean difference values for the sequential disability number groups (i.e., 1–2, 2–3, and 3–4+) are presented in Table 3. Overall, we found significant differences for all measures except SNAP Behavior Support Hours and SIS Exceptional Behavior Needs Total Score. With regard to the planned comparisons conducted to investigate the

Table 3. Mean Difference Values for Measures of Adaptive and Challenging Behavior and Support for Number of Disabilities

Measure ^a	Score range	Statistic ^b	Mean difference values for sequential disability number groups		
			1–2 disabilities	2–3 disabilities	3–4+ disabilities
Adaptive behavior					
ICAP Broad Independence ^c	328–540	16.84***	9.6	52.6**	5.3
Challenging behavior					
ICAP Generalized Maladaptive Behavior ^{d, f, g}	0–38	3.50*	–3.5	6.3**	–9.8**
SIS Exceptional Behavior Needs Total ^{d,e}	0–11	4.85	–10.5	13.2*	–10.2
SNAP Behavior Support Hours ^{d, f}	0–4.0	2.06	–0.3	0.6*	–0.2
Support needs					
ICAP Service Score ^c	16–96	14.85***	7.2	16.0**	11.6*
SIS Support Need Index ^c	63–133	14.25***	–3.5	–19.2**	–3.6
SNAP Day Support Hours ^c	0.3–10.2	9.43***	–1.0	–1.8**	–0.8
Medical/Health needs					
SIS Exceptional Medical Needs Total ^{c, e}	0–6	22.64***	–1.3	–16.7**	–13.2*
SNAP Physical Health Support Hours ^c	0–2.5	23.10***	–0.3*	–0.7**	–0.2

^aICAP = Inventory of Client and Agency Planning, SIS = Supports Intensity Scale, SNAP = Service Need Assessment Profile. ^bAll $F(3, 79)$ except boldface entries, which are $H(3)$. ^cOne-tailed test of significance. ^dTwo-tailed test of significance. ^eNonparametric comparisons (Kruskal-Wallis H test and Mann-Whitney U test), mean difference values reported are mean rank differences. ^fAnalyses conducted using transformed variable because distribution was not normally distributed, although the mean differences in the table relate to raw score mean differences. ^gAbsolute values used.

* $p < .05$. ** $p < .01$. *** $p < .001$.

impact of additional disabilities on skill levels and need for supports, the trends indicated above were further supported. The mean difference values revealed the trend of fewer adaptive behavior skills associated with increased number of disabilities present, although the only significant group difference found was between two and three disabilities for the ICAP Broad Independence Index. Mean difference values showed a greater need for overall and medical/health supports as the number of disabilities present increased. We found significant group differences between the presence of one and two disabilities (SNAP Physical Health Support Hours), two and three disabilities (all measures), and three and four or more disabilities (ICAP Service Score and SIS Exceptional Medical Needs Total Score). A number of significant differences between groups were found for the

challenging behavior measures; however, rather than a trend of more challenging behavior and a need for greater behavioral support associated with increased number of disabilities, the mean difference values and the score profiles in the top graph of Figure 3 reveal more complicated trends involving a number of directional changes.

To facilitate comparisons to determine the impact of cumulative severity, we formed four subgroups using the following groupings of the CSS, Group 1 = 1–2 ($n = 18$), Group 2 = 3–5 ($n = 26$), Group 3 = 6–8 ($n = 20$), and Group 4 = 9–14 ($n = 19$). Table 4 presents the results of these analyses and the mean difference values for planned comparisons involving the sequential CSS groups (i.e., Group 1–Group 2, Group 2–Group 3, etc).

The results indicate that significant mean differences existed between CSS groups for all

Table 4. Mean Difference Values for Sequential Cumulative Severity Groups for Measures of Adaptive and Challenging Behavior and Support

Measure ^a	Statistic ^b	Mean difference values for sequential cumulative severity groups		
		Group 1– Group 2	Group 2– Group 3	Group 3– Group 4
Adaptive behavior				
ICAP Broad Independence ^c	32.94***	21.1*	29.4**	47.4**
Challenging behavior				
ICAP Generalized Maladaptive Behavior ^{d,f,g}	1.91	−2.0	1.8	3.1
SIS Exceptional Behavior Needs Total ^{d,e}	2.27	−6.6	10.2	−5.3
SNAP Behavior Support Hours ^{d,f}	2.96*	−0.6	0.6*	0.1
Support needs				
ICAP Service Score ^c	19.53***	10.4*	10.8*	17.2**
SIS Support Need Index ^c	26.02***	−8.6*	−11.9**	−16.2**
SNAP Day Support Hours ^c	12.70***	−2.4**	−1.0	−0.8
Medical/Health needs				
SIS Exceptional Medical Needs Total ^{c,e}	16.15***	−10.7*	−7.4	−10.4
SNAP Physical Health Support Hours ^c	21.71***	−0.4**	−0.7**	−0.1

^aICAP = Inventory of Client and Agency Planning, SIS = Supports Intensity Scale, SNAP = Service Need Assessment Profile. ^bAll $F(3, 79)$ except boldface entries, which are $H(3)$ ^cOne-tailed test of significance. ^dTwo-tailed test of significance. ^eNonparametric comparisons (Kruskal-Wallis H test and Mann-Whitney U test), mean difference values reported are mean rank differences. ^fAnalyses conducted using transformed variable because distribution was not normally distributed, although the mean differences in the table relate to raw score mean differences. ^gAbsolute values used.

* $p < .05$. ** $p < .01$. *** $p < .001$.

measures except ICAP Generalized Maladaptive Behavior Index and SIS Exceptional Behavior Needs Total score. As with number of disabilities above, increased severity of overall disability resulted in fewer adaptive behavior skills present, as shown in the mean difference values for the sequential comparisons, with all comparisons between groups achieving statistical significance. Increased severity of total disability resulted in a profile of greater overall support required on all three measures, with statistically significant differences present between all sequential severity groups for the ICAP Service Score and the SIS Support Need Index. Only one sequential severity group difference was significant for SNAP Day Support Hours. Both measures of medical/health support revealed mean difference values reflecting greater needs associated with increased severity of disability, with three of the six sequential severity

group comparisons achieving significance. Only one significant finding occurred with the sequential group comparisons for the challenging behavior measures investigated. A trend involving a number of directional changes appeared to be present, based on the mean difference values for the challenging behavior measures, as evident in Figure 3 (bottom graph).

Discussion

Results showed a high incidence of multiple disabilities among a group of individuals with a primary intellectual disability. Consistent with the Australian Institute of Health and Welfare (2004) findings regarding comorbidity and disability severity, we found a significant association between number of disabilities and severity of intellectual disability. Further, higher prevalence rates for the

additional disability groups investigated and medical/health impairments were generally associated with more marked levels of intellectual impairment. An exception to this pattern, and one frequently reported in the literature, involves the prevalence of sensory disabilities, for which slightly lower prevalence rates were observed for individuals with severe/profound intellectual impairments. This finding may not reflect actual incidence but, rather, an underrecognition of sensory impairments, which has been described in the literature (Kerr et al., 2003; van Schrojenstein Lantman-de Valk, 2005). The lower incidence of psychiatric disabilities with increasing intellectual impairment is consistent with other studies; however, as with sensory impairments, it is unclear whether this is a prevalence issue or a recognition/diagnostic problem.

In addition to finding an association between level of intellectual impairment and number of disabilities present, we also found a meaningful association between the number of disabilities present and measures of adaptive behavior and support needs, with coefficients in the moderate range. The finding of moderate correlations between the number of disabilities present and medical support indicators highlights the importance of coexisting disabilities relating to the need for support in the area of medical/health issues. Notably, the findings of the comparisons conducted between the disability number groups indicated significantly higher need for medical/health supports associated with having more disabilities present. These findings have important implications for the assessment of support needs for the purposes of resource allocation, with medical/health needs an important consideration for individuals with intellectual and other coexisting disabilities.

Plotting mean scores indicated support for fewer adaptive behaviors and a need for more support as number of disabilities present increased, although the properties of these trends could not be established in this study using curve estimation. However, although the curve estimation statistics were not informative, the pattern of results for the comparisons between sequential disability number groups reported in Table 3 may provide an indication of the nature of the trends. Of particular note are the significantly lower skill levels and higher need for supports found for individuals with three disabilities when compared to skill and support levels for those with two disabilities.

Although the other comparisons reported in this table revealed similar trends, most differences were not significant. Thus, taken together, these findings indicate that the cumulative effect of additional disabilities on adaptive behaviors and need for supports for individuals with a primary intellectual disability is not best explained by an additive or steady growth trend. Rather, there may be some other trend, not examined in this study, which would explain the impact of the additional disabilities. Alternatively, the shapes of such trends may be complicated by the varying mix of additional disabilities present and the nature of the additional disability may be important in determining the actual profile of the trend, a possibility that warrants further investigation.

When considering the cumulative severity of disability measures employed in this study, we found that it is not only the number of disabilities present that impacts on adaptive behaviors and need for supports but also the severity of each additional disability. Correlations between measures of adaptive behavior and support needs (including medical needs) and the cumulative disability severity score were slightly stronger than number of disabilities on its own as an indicator of severity of disability. In addition, comparisons conducted between the severity groups revealed that adaptive behaviors and support needs, particularly as measured using the ICAP and SIS, were significantly impacted by the cumulative severity of disabilities. Again, curve estimation statistics did not clarify the nature of the growth/decline trends associated with increasing disability severity, with the small numbers of individuals at the higher levels of cumulative severity likely to be a limiting factor.

Throughout this study, we found an association between the number and severity of disabilities present and levels of adaptive behavior and intensity of needed supports (including medical/health supports), whereas this was not so for challenging behaviors or need for behavioral support. Graphing the profiles for challenging behaviors and need for behavioral support revealed a number of directional changes associated with the number and severity of additional disabilities present. We did not anticipate finding a trend of more challenging behaviors (and a need for more behavioral support) associated with the presence of multiple disabilities or more severe levels of total disability, as we predicted for adaptive behaviors and need for associated

supports. This expectation was supported by the lower correlations often reported between measures of challenging behavior and severity of intellectual impairment, as well as indications within the literature concerning associations between challenging behaviors and particular additional disabilities (e.g., epilepsy and psychiatric conditions), notwithstanding the fact that conflicting associations have frequently been reported. Thus, it is possible that the profiles for the challenging behaviors measures obtained in this study were influenced more by the nature of the additional disabilities present than by the number or severity of the additional disabilities.

Understanding the impact of the relationship between number of disabilities and severity of disabilities on adaptive functioning and support needs involves consideration of contemporary ecological models of disability, in which disability is understood in terms of a person–environment interaction. Although ecological models of disability are more recent conceptualizations, the importance of the person–environment interaction has been reported in the literature pertaining to aging for several decades. Lawton and Nahemow's (1973) ecological model of aging is one such model. For this model it is proposed that an individual's health and well-being are a function of balance between personal competence and environmental demands and that for each level of competence, there are different ranges of environmental demand at which an individual is able to function effectively (Moore, VanHaitsma, Curyto, & Saperstein, 2003). Those with high personal competence are considered to have a broader scope for interacting and adapting to their environment in ways that maximize positive behaviors and outcomes, whereas those who are less competent are more vulnerable to the demands of the environment (Lawton & Nahemow, 1973) and, consequently, would require higher levels of environmental support (Stark, 2001).

The application of supports to lessen the discrepancy between the person's capacity and the environmental demands as well as to ameliorate the experience of disability is the foundation of contemporary models of disability such as the *ICF* (World Health Organization, 2001) and the American Association on Intellectual and Developmental Disabilities (AAIDD) system of supports classification (Luckasson et al., 2002). In the present study, significant reductions in functional

capacity (measured with an adaptive behavior scale) were found for individuals with three disabilities compared to those having two disabilities and as the severity of disabilities increased. In terms of this ecological model, the reduced functional capacity would be considered to lessen the individual's threshold for coping with environmental demands, thereby necessitating the introduction of supports aimed at improving individual functioning.

Further, in most cases the need for significantly higher levels of support was correspondingly identified. Exceptions include the significantly higher level of supports needed by individuals with four or more disabilities than those with three disabilities, identified by the ICAP Service Score. This finding is likely to be associated with the method used for computing the Service Score, in which both adaptive and challenging behaviors have been included. Consequently, the finding of a significant difference for these groups may be related to the significantly higher level of challenging behavior identified on the ICAP Generalized Maladaptive measure for those with four or more disabilities, but which was not found to be significant on the other two challenging behavior measures considered.

In addition, the profile of results for SNAP Day Support Hours differed from the findings for the ICAP and SIS with respect to the cumulative severity of disabilities comparisons. Again, scale composition may be important. Of the three scales used, SNAP differs from the other scales because it incorporates measures of behavioral and health supports within the total score used in these analyses. The inclusion of these dimensions of support within the total score may contribute to the pattern of results obtained.

Notwithstanding these exceptions, current findings highlight the importance of considering additional disabilities for individuals with a primary intellectual disability. The findings have demonstrated coexisting disabilities to have an impact on functional capacity and consequent need for supports that are clinically significant. Further, the severity of disabilities has also been demonstrated to impact skill levels and consequent need for supports. This correspondence between level of functioning and need for support is assumed in the AAIDD intellectual disability classification system, in which severity levels have been replaced by support intensity levels (Leonard

& Wen, 2002). Supporting this correspondence between adaptive behaviors and need for supports are the high correlations obtained between measures of adaptive behavior and measures of support needs previously reported in the literature (Harries, Guscia, Kirby, Nettelbeck, & Taplin, 2005; Severance, 2007).

The findings of this study should be considered in terms of previous research findings and possible limitations. First, the number of disabilities reported was established using caregiver reports rather than screening tools or clinical assessment. However, sizeable discrepancies and underreporting have been demonstrated using this approach (Carvill, 2001; Kerr et al., 2003; Vlas-kamp, 2005; Warburg, 2001). Consequently, the actual number of disabilities present may have been higher than reported here, which would impact on analyses using both the number of disabilities and also the CSS. Nonetheless, the patterns of results do indicate that not only are the number of additional disabilities present likely to impact on skill levels and support needs, but the severity of these additional disabilities will also.

An additional limitation is the method of determining both the severity of the disabilities and the relatively crude and exploratory method used for deriving the cumulative severity measure. Severity of each disability was established using caregiver estimates instead of an objective measure of disability severity, due to a lack of standardized and universal measures that could be used across the range of disabilities considered here. In future studies the generic scale of the *ICF* (World Health Organization, 2001), which incorporates qualifying words and percentages to quantify level of impairment, may ensure higher reliability for these ratings, which could then be used to investigate alternative methods for deriving a cumulative severity score of this kind.

Third, the study group was not large and, consequently, there were small numbers in the higher disability number and cumulative severity groups, which may have contributed to difficulties determining trends in the curve estimation analyses. Finally, the participants were all individuals receiving support from residential services and, as previously noted, represented a group of individuals with generally higher incidences of additional disabilities than those of the overall population of individuals with intellectual disabilities who receive support services from all government-funded disability support services.

Although levels of adaptive and challenging behavior and consequent support needs for participants may differ from individuals not residing in such facilities, it is, nonetheless, considered important to understand the link between the presence of additional disabilities and need for support for this group of individuals. A large proportion of Australian government funding of disability services is directed towards the provision of accommodation supports (accounting for about half of total expenditure), with similar levels of expenditure (54%) reported from the United Kingdom for accommodation/residential support (Australian Institute of Health and Welfare, 2005; Felce et al., 2008). Fifty three percent of recipients of accommodation support services in Australia are reported to have a primary diagnosis of intellectual disability (Australian Institute of Health and Welfare, 2007), making the allocation of support resources to the group represented in this study a substantial concern for government bodies.

Notwithstanding these limitations, the findings of this study highlight the impact of the number and severity of additional disabilities on need for support for individuals with a primary intellectual disability. As previously indicated, it is possible that some individuals in this study had undetected impairments and that these impairments impact on quality of life and capacity to undertake activities of daily living. Identification of such impairments could result in the application of appropriate supports or environmental modifications that could ameliorate the disadvantage experienced. The widely noted underreporting of health and impairment conditions has led to recommendations of regular health checks, particularly to identify the onset of impairments associated with aging (Janicki & Dalton, 1998; Kerr et al., 2003) or the use of a health surveillance protocol, as described by Traci et al. (2002). Knowledge about the additional impairments present would also allow the identification of an appropriately skilled multidisciplinary support team because the nature and intensity of supports required would be impacted not only by the number and severity of disabilities present, but also by the nature of the additional disabilities. Further research is needed to investigate the importance of the nature and severity of additional disabilities present in terms of the impact on adaptive and challenging behaviors and need for support.

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